# ORGAN SYSTEM ANATOMY (POST LEC)

## Lecture #1: Introduction to Anatomy | OPHT 1-1 | PRELIMS

*By Accel Jimenez*

## ANATOMY AND PHYSIOLOGY

- ★ Anatomy
  - scientific discipline that investigates the body's structures. (shape/size of the bones).
  - examines the relationship between the structure of a body part.

*Anatomy studied in different levels:*

- ➢ **Developmental Anatomy** - studies the structural changes that occur between conception and adulthood.
- ➢ **Microscopic Anatomy** - Study that focuses on cells and tissues, revealing their organization and function within the body.
  - Embryology - changes from conception to the end of the eighth week of development.
  - Cytology - examines the structural features of cells.
  - Histology - examines tissues (composed of cells and materials surrounding them).
- ➢ **Macroscopic Anatomy-** study of structures that can be examined without the aid of a microscope, can be approached either systemically or regionally (Gross Anatomy.)
- ➢ **Surface Anatomy** - involves looking at the exterior of the body to visualize structures deeper inside the body.

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|---|
| Systemic Anatomy - system by system<br>**Regional Anatomy** - area by area (head, abdomen, arm) |

- ★ **Physiology**
  - the scientific investigation of the processes or functions of living things.
  - Often examines systems rather than regions because a particular function can involve portions of a system in more than one region.
  - Studies:
    - ➢ Muscle contraction
    - ➢ Nerve impulses
    - ➢ Gas exchange
    - ➢ Circulation of blood

Major goals when studying human physiology:

1. examining the body's responses to stimuli
2. examining the body's maintenance of stable internal conditions within a narrow range of values in a constantly changing environment.

*Physiology studied in different levels:*

- ➢ **Cell physiology** - examines the processes occurring in cells such as energy production from food.
- ➢ **Systemic physiology** - considers the functions of organ systems.
  - Cardiovascular physiology
  - Neurophysiology
- ➢ **Pathology** - medical science dealing with all aspects of disease, with an emphasis on the cause and development of abnormal conditions, as well as the structural and functional changes resulting from disease.

## STRUCTURAL/FUNCTIONAL ORGANIZATION OF THE HUMAN BODY

1. **Chemical level**
   - ■ involves how atoms, such as hydrogen and carbon, interact and combine to form molecules.
   - ■ Molecules' structure determines its function.
2. **Cell level**
   - ■ basic structural and functional units of all living organisms.
   - ■ (organelles) A structures inside cells carry out particular functions, i.e., nucleus.
3. **Tissue level**
   - ■ composed of a group of similar cells and the materials surrounding them

*4 types of tissue:*

- ■ Epithelial Tissue
  - ☑ covering/lining of the body (skin).
- ■ Connective Tissue
  - ☑ Most abundant tissue in the body.
  - ☑ Facilitates blood flow.
- ■ Muscle Tissue
  - ☑ Moves and supports your organs.

- Nervous Tissue
  - ☑ Responsible for transmitting signals throughout the body. Controls bodily functions.
  - ☑ 2 types: Neurons (nerve impulses) & Glial Cells (supports neurons)

4. **Organ level**
   - composed of two or more tissue types that perform one or more common functions. (bladder, stomach, lungs)
5. **Organ System level**
   - a group of organs that together perform a common function or set of functions and are therefore viewed as a unit.

*There are 12 Organ systems:*

- Integumentary
- Skeletal
- Cardiovascular
- Muscular
- Respiratory
- Reproductive
- Nervous
  - ➢ Sensory receptors (always upwards) Also modulates breathing patterns.
- Endocrine
  - ➢ hormones
- Lymphatic
  - ➢ Cleans excess blood (filtration system)
  - ➢ Females have more nodes
  - ➢ A female's immune system weakens when in the menopause stage.
- Digestive
  - ➢ Digestion starts in the tongue.
- Urinary
  - ➢ Most important part: Kidney

6. **Organism Level**
   - any living thing considered as a whole—whether composed of one cell, such as a bacterium, or of trillions of cells, such as a human.

**FUNCTIONAL UNIT = MAJOR CELL**
Cell = Cyte
To get the functional unit:
1. Medical term of the organ
2. Add -cyte

## CHARACTERISTICS OF LIFE

★ **Organization**
- specific interrelationships among the parts of an organism.
- Ex. cell, tissue, organ,

★ **Metabolism**
- ability to use energy and to perform other vital functions.
- refers to all of the chemical reactions taking place in the cells and internal environment of an organism.
  - Catabolism - breaks down molecules into simpler ones. (large to small)
  - Anabolism - builds complex molecules from simpler ones. (small to large)

★ **Responsiveness**
- an organism's ability to sense changes in its external or internal environment and adjust to those changes.

The nervous and endocrine systems regulate responses to changes in the environment through **cell-to-cell communication**

★ **Growth**
- increase in the size or number of cells, which produces an overall enlargement of all or part of an organism.

Increase in size = Hypertrophy
Increase in cells = Hyperplasia

★ **Reproduction**
- formation of new cells or new organisms.
- allows for growth and development.
- allows all living organisms to pass on their genes to their offspring.

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Functional Unit of Bones
- Osteocyte (old)
- Haversian system (new)

★ **Development**
- the changes an organism undergoes through time, beginning with fertilization and ending at death.
- **Greatest developmental changes** occur BEFORE birth.
- **Differentiation**: *changes in a cell's structure and function* from an immature, generalized state to a mature, specialized state.

➢ **Morphogenesis**: change in shape of tissues, organs, and the entire organism.

## HOMEOSTASIS

- the existence and maintenance of a relatively constant environment within the body. As our bodies undergo their everyday processes, we are continuously exposed to new conditions. Changes in our external environmental conditions can result in changes in our internal body conditions.

**Variables** - Changes in internal body conditions because their values are not constant.

★ Feedback Loops
- Regulates homeostasis.

★ Negative Feedback
- The response by the effector is stopped once the variable returns to its set point.
- Most abundant process.

★ Positive Feedback
- occur when a response to the original stimulus results in the deviation from the set point becoming even greater.

- Negative feedback
  **HIGH = NF WILL DECREASE**
  ➢ Ex. bloodclotting
- Positive Feedback
  **HIGH = PF WILL INCREASE MORE**
  ➢ Ex. child birth

*Two basic principle about homeostasis mechanisms:*
1. Many disease states result from the failure of negative-feedback mechanisms to maintain homeostasis.
2. Some positive-feedback mechanisms can be detrimental instead of helpful.

## BASIC ANATOMICAL TERMONOLOGY

- **Anatomical Position**
  - refers to a person standing erect with the face directed forward, the upper limbs hanging to the sides, and the palms of the *hands facing forward.*
  - Frame of reference
- **Fundamental Position**
  - Palm facing lateral (side)

| | |
|---|---|
| Supine | Lying face upward |
| Prone | Lying face downward |
| Fowler's Position | Inclined |
| Trendelenburg Position | Declined |
| Lateral Recumbent Position (Right/Left) | Lying to the side |

## DIRECTIONAL TERMS

| | |
|---|---|
| Anterior (Ventral) | Front of the face |
| Posterior (Dorsal) | Back part |
| Superior (Cranial) | Upper Part |
| Inferior (Caudal) | Lower Part |
| Medial | Toward the midline of the body |
| Lateral | Toward the outside of the body. |
| Proximal | Applicable only on limbs, higher |
| Distal | Applicable only on limbs, lower |
| Superficial (Internal) | Outer |
| Deep (External) | Inner |

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LIMBS = EXTREMITY
- Upper Extremity - arms
- Lower Extremity - legs

Elbow is **DISTAL** to shoulders
distal proximal

Elbow is **PROXIMAL** to wrist
proximal distal

## PLANES

- **Sagittal Plane:** Divides body into right & left.
- **Median Plane:** Equal middle right & left halves.
- **Transverse Plane:** Divides into top (superior) & bottom (inferior).
- **Frontal (Coronal) Plane:** Divides into front (anterior) & back (posterior).
    - Example: Coronal suture is where a crown would sit.

*Axis (point where it pierces through):*

| | |
|---|---|
| X-Axis | Pierces through lateral to medial (Horizontal Line) |
| Y-Axis | Pierces through anterior to posterior |
| Z-Axis | Pierces through superior to inferior (Vertical Line) |

| TYPE | PLANE | AXIS | MOTIONS |
|---|---|---|---|
| Coronal / Frontal Plane | XZ Plane | Y-Axis | Abduction<br>Adduction<br>Lateral flexion<br>Radial deviation<br>Unar Deviation |
| Sagittal / Median Plane | YZ Plane | X-Axis | Flexion<br>Extension<br>Forward flexion<br>Dorsiflexion<br>Plantarflexion |
| Transverse Plane | XY Plane | Z-Axis | Exterior Rotation<br>Interior Rotation<br>Lateral Rotation<br>Spine & shoulders only |

| Type | Ref Point | Motion |
|---|---|---|
| Hip w Knee Extended | Toes | Ext. Rotation<br>Int Rotation |
| Hip w Knee Flex | Knee | |